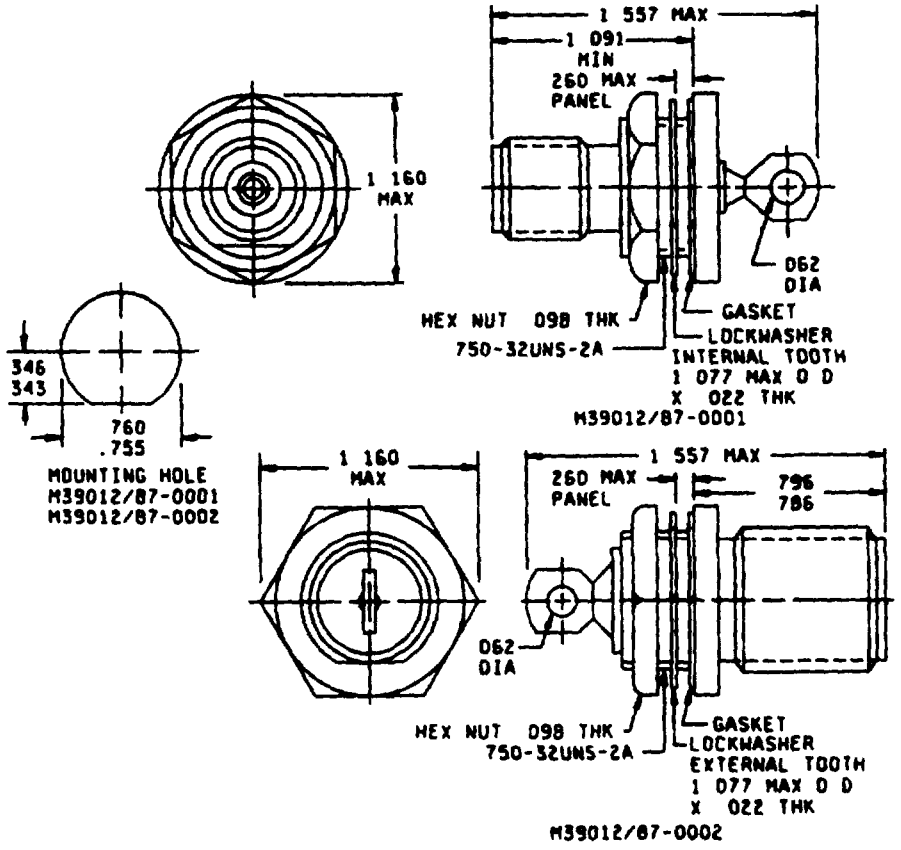
NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

PERFORMANCE SPECIFICATION

CONNECTORS, COAXIAL, RADIO FREQUENCY,
SERIES QSC (UNCABLED - RECEPTACLE, SOCKET CONTACT, HERMETIC SEALED,
JAM NUT, FRONT AND REAR MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-PRF-39012



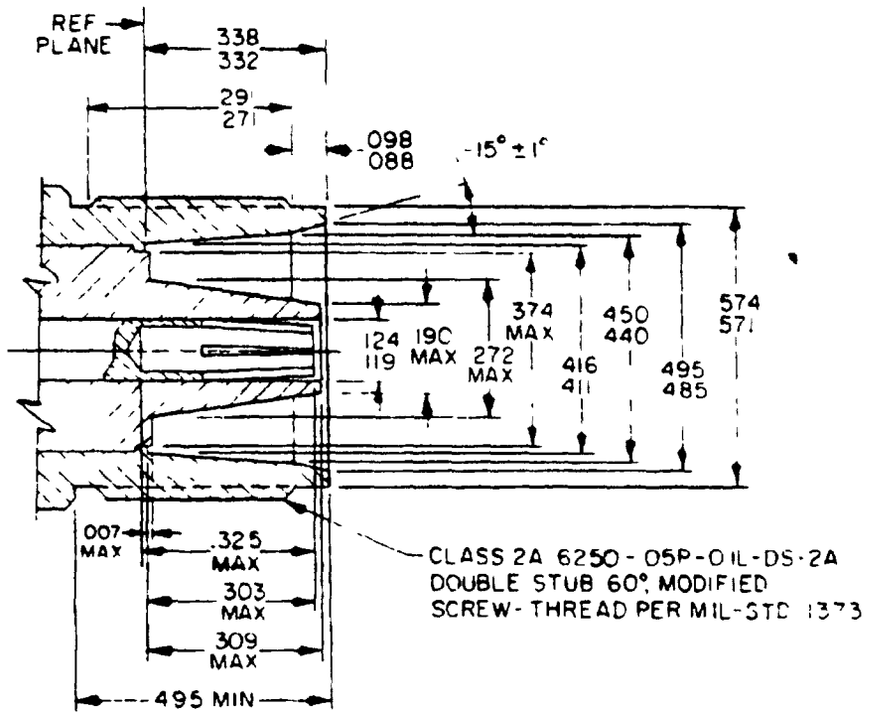
INCHES	MM
022	56
062	1 57
098	2 50
260	6 60
343	8 71
346	8 79
750	19 05
755	19 18
760	19 30
786	19 96
796	20 22
1 077	27 36
1 091	27 71
1 169	29 46
1 557	39 55

NOTES
1 Dimensions are in inches
2 Dimension 1 160 defines the overall diameter
3 Dimension 1 557 defines the overall length
4 Metric equivalents (to the nearest 01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
5 Wrench flats to accomodate standard wrench opening per H-28, app 10

FIGURE 1 General configuration

FSC 5935

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INCHES	MM	INCHES	MM	INCHES	MM
007	18	291	7 39	416	10 56
088	2.24	.303	7 70	440	11 18
098	2.50	309	7 85	450	11 43
119	3.02	.325	8.26	.485	12.32
124	3 15	.332	8.43	495	12.57
.190	4.83	.338	8 59	571	14 50
.271	6.88	.374	9.50	.574	14 58
.272	6.91	411	10.44		

NOTES
1 Dimensions are in inches.
2 Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm

FIGURE 2 Mating dimensions for socket terminations

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ENGINEERING DATA

Nominal impedance 50 ohms

Frequency range. 0 to 11,000 MHz

Voltage rating 1,500 volts rms maximum working voltage at sea level 325 volts rms maximum at 70 000 feet

Temperature rating -65° to +165°C.

REQUIREMENTS

Dimensions and configuration See figure 1

Force to engage and disengage
Longitudinal force - Not applicable.
Torque - 2-1/2 inch-pounds maximum

Coupling proof torque
15 inch-pounds minimum

Inspection conditions 6-10 inch-pounds

Mating characteristics
See figure 2 for dimensions
Center contact (socket)
Oversize test pin - .098 diameter minimum (nonclosed entry contacts only)
Insertion depth - 125 minimum
Number of insertions - One

Insertion force test - Steel test pin dia .092 minimum
Test pin finish - 16 microinches
Insertion force - 2 pounds maximum

Withdrawal force test· Steel test pin dia .090 maximum
Withdrawal force - 2 ounces minimum.
Test pin finish - 16 microinches

Hermetic seal. Leakage shall not exceed $1x10^{-5}$ cc/sec of tracer gas at atmospheric pressure

Leakage (pressurized connectors)
Not applicable.

Insulation resistance Method 302, test condition B MIL-STD-202 5,000 megohms minimum

Center contact retention
6 pounds minimum axial force

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR)
Not applicable.

Swept frequency VSWR test setup
Not applicable

Connector durability
500 cycles minimum at 12 cycles/min maximum
The mating force shall not diminish by more than 10 percent

Contact resistance In milliohms maximum

	Initial	After environment
Center contact	1.0	1.5
Outer contact	20	Not applicable

Dielectric withstanding voltage· Method 301 of MIL-STD-202 4,000 volts rms at sea level.

Vibration, high frequency Method 204, test condition B, MIL-STD-202

Shock Method 213, test condition G, MIL-STD-202

Temperature cycling Method 102, test condition C, MIL-STD-202

Thermal shock Not applicable

Moisture resistance Method 106 of MIL-STD-202 No measurements at high humidity Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity

Corona level Not applicable

RF high potential withstanding voltage
Voltage and frequency - 2,500 volts rms at 3 MHz
Leakage current - Not applicable

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Cable retention force. Not applicable

Coupling mechanism retention force:
Not applicable.

RF leakage. Not applicable.

Insertion loss.
Not applicable.

Qualification not required.

Part number. M39012/67 -0001 for the front and M39012/67 -0002 for the rear-mounted connector.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI, EL
DSA - ES

User activities.
Army - AT, AV, ME

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5935-1862-4)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-C-39012/87 CONNECTORS, COAXIAL, RADIO FREQUENCY, SERIES QSC (UNCABLED-RECEPTACLE, SOCKET CONTACT, HERMETIC SEALED, JAM NUT, FRONT

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT ___ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

___ YES ___ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity*

SUBMITTED BY *(Printed or typed name and activity) Optional*

DATE

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